$=50+90\cos(100t-90^\circ)\ V$

6-1. 如图 1 所示电路，$u_s=50+90\sin(100t)$V，$i_s=20\cos(50t)$A。试计算：（1）稳态响应 i_L、u_C；（2）电压源提供的有功功率。

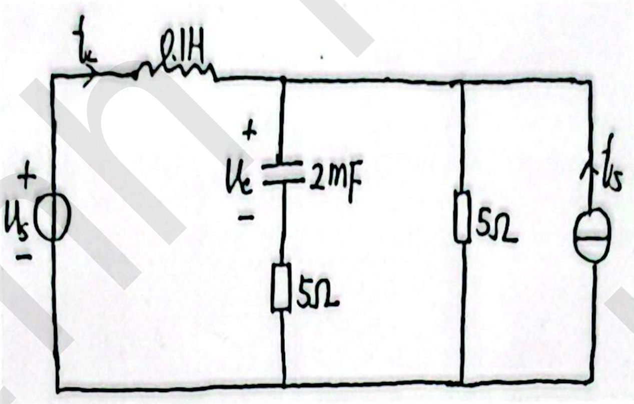
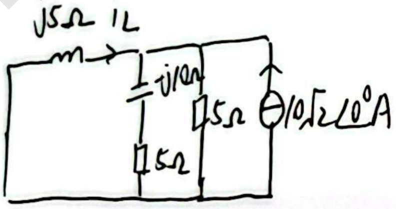


图 1

(1) ①直流分量作用，$I_{L0}=\frac{50}{5}=10A$，$V_{C0}=50V$

②基波分量作用时，$X_L=5\Omega$，$X_C=\frac{1}{wC}=10\Omega$



$$\dot{I}_{L1}=-\frac{\frac{1}{j5}}{\frac{1}{5}+\frac{1}{5-j10}+\frac{1}{j5}}\times10\sqrt{2}\angle0^\circ=10.5538\angle116.57^\circ A\ (分流公式)$$

$$\dot{U}_{C1}=\frac{\frac{1}{5-j10}}{\frac{1}{5}+\frac{1}{5-j10}+\frac{1}{j5}}\cdot(-j10)\cdot10\sqrt{2}\angle0^\circ=47.19\angle0^\circ V$$

③ $w=100rad/s$时，$X_L=10\Omega$，$X_C=5\Omega$

$$\dot{I}_{L2}=\frac{45\sqrt{2}\angle-90^\circ}{j10+\frac{(5-j5)\times5}{10-j5}}=\frac{45\sqrt{2}\angle-90^\circ}{3+j9}=6.7082\angle-161.57^\circ A$$

$$\dot{U}_{C2}=6.7082\angle-161.57^\circ\times\frac{5}{10-j5}\times(-j5)=15\angle135^\circ V\ (将\ \dot{I}_{L2}分流再乘以容抗)$$

$$i_L=[10+14.93\cos(50t+116.57^\circ)+9.49\cos(100t-161.57^\circ)]A$$

$$u_C=[50+66.74\cos(50t)+21.21\cos(100t+135^\circ)]V$$

(2) $$P=50\times10+\frac{9.49\times90}{2}\cos(-90^\circ+161.57^\circ)=635W$$

6-2. 如图 2 所示电路，已知

$R = 10\Omega, \omega L = 2\Omega, \dfrac{1}{\omega C} = 18\Omega, u_s = 10 + 8\sqrt{2}\cos(\omega t) + 12\sqrt{2}\cos(3\omega t + 30)\text{V}$,

$i_s = 5\sqrt{2}\cos(\omega t + 60)\text{A}$.求电流表的读数以及两电源各自发出的功率

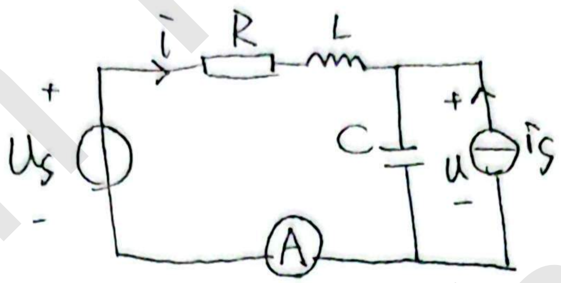


图 2

①直流作用，$I_0 = 0A$，$U_0 = 10V$

②基波单独作用，由叠加定理有，U_s和i_s分别单独作用

$\dot{I}_1 = \dfrac{8\angle 0^\circ}{10 - j16} - \dfrac{-j18}{10 - j16} \times 5\angle 60^\circ = 4.4079\angle -154.77^\circ A$

$\dot{U}_1 = (\dot{I}_1 + \dot{I}_s) \times (-j18) = (4.4079\angle -154.77^\circ + 5\angle 60^\circ) \times 18\angle -90^\circ = 51.61\angle 31.25^\circ V$

③三次谐波作用时，$\dot{I}_3 = \dfrac{12\angle 30^\circ}{10} = 1.2\angle 30^\circ A$

读数 $I = \sqrt{I_1^2 + I_3^2} = 4.568A$

$P_{us} = 8 \times 4.4079 \times \cos(154.77^\circ) + 1.2 \times 12 = -17.5W$

$P_{is} = 51.61 \times 5 \times \cos(31.25^\circ - 60^\circ) = 226.24W$

7-1. 如图 3 所示的正弦交流稳态电路，M=10mH，L_1=10mH，L_2=40mH，C=4/3μF。已知电压源有效值 U_s=50V。试求：电压源频率为多大时电流表的读数最小，并求此最小值。

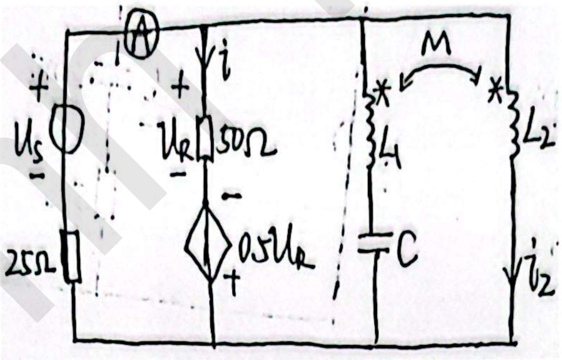
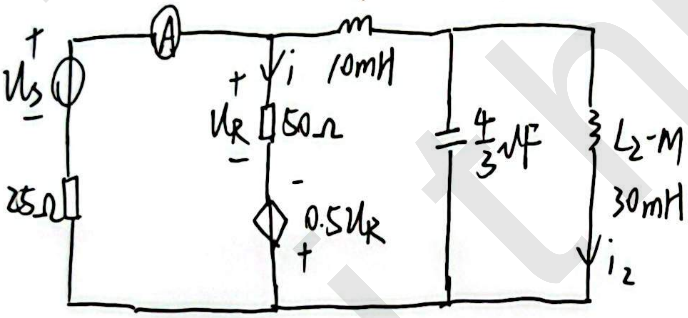


图 3

右侧并联谐振时，电流表读数最小



$$W(L_2-M)=\frac{1}{WC},\quad W=\sqrt{\frac{1}{(L_2-M)C}}=5000rad/s,\quad f=\frac{W}{2\pi}=795.77Hz$$

$$\begin{cases} I=\dfrac{50+0.5U_R}{75} \\ U_R=50I \end{cases} \Rightarrow I=1A$$

7-2．如图 4 所示电路处在谐振状态，$R_2 = 3\omega L = 2\frac{1}{\omega C}, R_1 = 1\Omega, U_S = 20V$，电流表 A_1 的读数是 30A， 求电流表 A_2 和功率表 W 的读节，并求电容 C_2 的容抗

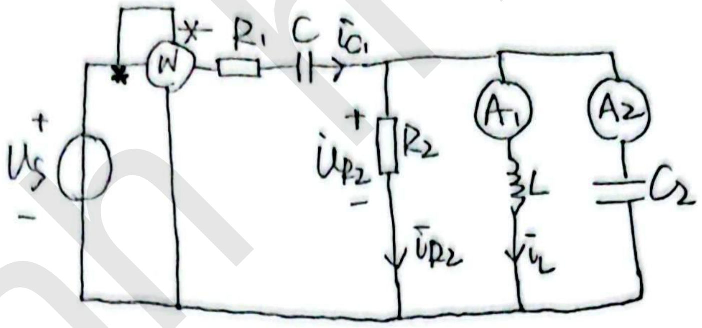


图 4

电感L支路. $\dot I_L = \frac{\dot U_{R2}}{j\omega L} = -j3\frac{\dot U_{R2}}{R_2} = -j3\dot I_{R2}$

$I_L = 30A$ ∴ $I_{R2} = 10A$ 设 $\dot I_{R2} = 10\angle 0°$ $\dot I_L = 30\angle -90° A$

$\dot I_{C2} = I_{C2}\angle 90°$ $\dot I_{C1} = 10 + j(I_{C2} - 30)$

等效阻抗: $Z = R_1 + \frac{1}{j\omega C} + \frac{\dot U_{R2}}{\dot I_{C1}}$

$= R_1 - j\frac{R_2}{2} + \frac{10\cdot R_2}{10 + j(I_{C2} - 30)}$

$= R_1 + \frac{100R_2}{100 + (I_{C2} - 30)^2} - j\left(\frac{R_2}{2} + \frac{10R_2(I_{C2} - 30)}{100 + (I_{C2} - 30)^2}\right)$

∴ $\frac{R_2}{2} + \frac{10R_2(I_{C2} - 30)}{100 + (I_{C2} - 30)^2} = 0$ ∴ $I_{C2} = 20A$

$\dot I_{C2} = 20\angle 90°$ $\dot I_{C1} = 10 - j10 = 10\sqrt{2}\angle -45° A$

$\dot U_S = (10 - j10)(1 + \frac{1}{j\omega C}) + 10R_2$ $U_S = 20V$ $R_2 = (2\sqrt{2} - 2)\Omega$

$P = I_{C1}^2 R_1 + I_{R2}^2 R_2 = 28.28W$

$X_{C2} = \frac{I_{R2}\cdot R_2}{I_{C2}} = 0.414\Omega$